Dazhong Shen

University of Science and Technology of China
Building 2, Institute of Advanced Technology,
Hefei, Anhui, P.R.China, 230000

Mobile Phone: +86 15256530521
Email: sdz@mail.ustc.edu.cn
Homepage: www.shendazhong.xyz

RESEARCH INTERESTS

My general research interest includes data mining and machine learning, focusing on generative models, variational inference, graph neural networks, and their data intensive applications.

EDUCATION

Ph.D. Sep. 2017 $\sim$ *Expected:* June 2023
Computer Science, University of Science and Technology of China (USTC)
Advisor: Prof. Hui Xiong, AAAS/IEEE Fellow
Major: Computer Science in Application

B.S. Sep. 2013 $\sim$ Jul. 2017
School of Mathematical Sciences, University of Science and Technology of China (USTC)
Major: Computational Mathematics, GPA: 3.63/4.30 (Major 3.82/4.30)

PROFESSIONAL EXPERIENCE

Research Intern, Talent Intelligence Center, Baidu Inc.
Mar. 2017 $\sim$ Sep. 2017 and Aug. 2018 $\sim$ June 2022, Mentor: Dr. Hengshu Zhu

- Developed a novel neural topic model with Graph Neural Networks by transforming each document into a directed graph with word dependency as edges, where **a well-defined generative story for the document graph is designed**, and **the topic can be visualized in a graph manner**. The related paper has been published on NeurIPS-2021.
- Proposed a simple but effective and generalized strategy to avoid the posterior collapse in variational autoencoder by collecting the geometric properties of latent space, i.e., diversity and uncertainty, where **Batch Normalization and Dropout are applied on the posterior's parameters**. The related paper has been published on IJCAI-2021.
- Developed two topic models (traditional and neural) to jointly represent multiple textual collects, where both hierarchical relationships among them and individual characteristics were modeled in a probabilistic manner. **The algorithm is launched in Baidu campus recruitment system to enhance talent searching, person-job fitting, and interview question recommendation**. Related papers have been published on IJCAI-2018 and ACM TOIS.
- Designed a novel Graph Convolution Nerual with the maximization operation to **handle the over-smoothing and over-squashing problems** simultaneously. The related paper was submitted to NeurIPS-2022

Data Contest, Aftershock Detection Contest, Directed by China Earthquake Administration
May 2017 $\sim$ Oct. 2017

- Developed an ensemble learning-based framework for identifying seismic P-phase arrivals automatically of the aftershocks. Based on our great performance, we won **the championship in the First Season and the runner-up in the Finals from 1,143 teams participated from around the world**. The related work was also published on SCIENCE CHINA Information Sciences (Chinese).

Research Assistant, The Hong Kong University of Science and Technology (Guangzhou)
Oct. 2022 $\sim$ Present, Advisor: Prof. Hui Xiong

- Plan to do some interesting research works about graph neural network, generative models, or causal inference.

AWARDS

- China National Scholarship. (for PhD students) 2022
- Huawei Scholarship (USTC). (for postgraduate and PhD students) 2018, 2021
- Global Digital Technology Scholarship. 2018
- IJCAI-2018 Student Travel Grants. 2018
- Aftershock Detection Contest (China Earthquake Administration), Runner-Up. 2017
- Academic Scholarship (USTC), Grad 1. five times from 2017 to 2021
- Zhang Zongzhi Science and Technology Scholarship (USTC). 2016
- Outstanding Student Scholarship (USTC), Silver Medal. 2015
- Outstanding Student Scholarship (USTC), Bronze Medal. 2014

PUBLICATIONS (7 CCF A, 1 CCF A Chinese Journal)

1. **Dazhong Shen**, Chuan Qin, Hengshu Zhu, Tong Xu, Enhong Chen, Hui Xiong. Joint Representation Learning with Relation-enhanced Topic Models for Intelligent Job Interview Assessment, ACM Transactions on Information Systems **(ACM TOIS)**, 2021. **(CCF A)**
2. **Dazhong Shen**, Chuan Qin, Chao Wang, Zheng Dong, Hengshu Zhu, Hui Xiong. Topic Modeling Revisited: A Document Graph-based Neural Network Perspective. The 35th Conference on Neural Information Processing Systems **(NeurIPS-2021)**, 2021. **(CCF A)**
3. **Dazhong Shen**, Chuan Qin, Chao Wang, Hengshu Zhu, Enhong Chen, Hui Xiong. Regularizing Variational Autoencoder with Diversity and Uncertainty Awareness. The 30th International Joint Conference on Artificial Intelligence **(IJCAI-2021)**, 2021. **(CCF A)**
4. **Dazhong Shen**, Hengshu Zhu, Chen Zhu, Tong Xu, Chao Ma, Hui Xiong. A Joint Learning Approach to Intelligent Job Interview Assessment, The 27th International Joint Conference on Artificial Intelligence **(IJCAI-2018)**, 2018. **(CCF A)**
5. 申大忠, 张琦, 徐童, 祝恒书, 赵雯佳, 殷子凯, 周培伦, 房立华, 陈恩红, 熊辉. EL-Picker：基于集成学习的余震P波初动实时拾取方法, 《中国科学: 信息科学》, 2020. **(CCF A Chinese)**
6. Qingxin Meng, Keli Xiao, **Dazhong Shen**, Hengshu Zhu, Hui Xiong, Fine-Grained Job Salary Benchmarking with a Nonparametric Dirichlet-process-based Latent Factor Model. INFORMS Journal on Computing **(JOC)**, 2022. **(UTD-24)**
7. Chuan Qin, Kaichun Yao, Hengshu Zhu, Tong Xu, **Dazhong Shen**, Enhong Chen, Hui Xiong. Towards Automatic Job Description Generation with Capability-Aware Neural Networks, IEEE Transactions on Knowledge and Data Engineering **(IEEE TKDE)**, 2022. **(CCF A)**
8. Zhi Zheng, Chao Wang, Tong Xu, **Dazhong Shen**, Penggang Qin, Xiangyu Zhao, Baoxing Huai, Xian Wu, Enhong Chen. Interaction-aware Drug Package Recommendation via Policy Gradient. ACM Transactions on Information Systems **(ACM TOIS)**, 2022. **(CCF A)**

9. Zhi Zheng, Chao Wang, Tong Xu, **Dazhong Shen**, Penggang Qin, Baoxing Huai, Tongzhu Liu, Enhong Chen. Drug Package Recommendation via Interaction-aware Graph Induction. The 30th International World Wide Web Conference **(WWW-2021)**, 2021. **(CCF A)**

Under Review:

10. **Dazhong Shen**, et al. Handling Over-Smoothing and Over-Squashing in Graph Convolution with Maximization Operation. IEEE Transactions on Neural Networks and Learning Systems (TNNLS), Submitted.
11. **Dazhong Shen**, et al. Exploiting Connections among Personality, Job Position, and Work Behavior: Evidence from Joint Bayesian Learning. ACM Transactions on Management Information Systems (TMIS), Submitted.

PATENTS

1. 申大忠，祝恒书，马超，朱琛，熊辉. 基于人工智能的辅助面试方法、装置及存储介质：中国，CN110472647[A]. 2019-11-19.
2. 申大忠，秦川，王超，董政，祝恒书，熊辉. 文本主题的确定方法、装置及电子设备: 中国，CN114492393[A]. 2022-05-13.
3. 秦川，姚开春，祝恒书，马超，申大忠. 一种招聘岗位描述文本的生成方法、装置、设备及介质: 中国，CN113627135[A]. 2021-11-09.
4. 苏昱涵，秦川，申大忠，赵洪科，宋欣，祝恒书. 用于处理技能信息的方法、模型训练方法及装置: 中国，CN114330333[A]. 2022-04-12.
5. 查睿，秦川，张乐，申大忠，徐童，祝恒书. 职业流动关系的预测方法、系统、设备及存储介质: 中国，CN114331380[A]. 2022-04-12
6. Chuan Qin, Kaichun Yao, Hengshu Zhu, Chao Ma, **Dazhong Shen**. Method, apparatus, device and medium for generating recruitment position description text: US, US20210216726[A]. 2021-07-15.

ACADEMIC SERVICES

Program Committees Member:

- The 39th International Conference on Machine Learning (ICML-2022)
- The 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD-2022)
- The 36th Conference on Neural Information Processing Systems (NeurIPS-2022)
- The 37th AAAI Conference on Artificial Intelligence (AAAI-2023)

Sub-reviewer for Journals and Conferences:

- IEEE TKDE, ACM TOIS, IEEE TSMC, etc.
- ICML, NeurIPS, KDD, WWW, SIGIR, AAAI, IJCAI, CVPR, ICDM, CIKM, WSDM, etc.